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Seeing the Forest for the Trees: the Lyman-alpha Forest as a Cosmological Signal for the Vera Rubin Observatory

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ABSTRACT

The scattering of light by the Lyman-α transition of neutral hydrogen in the intergalactic medium produces the so-called "Lyman-α Forest", visible in the observed spectra of distant quasars. This same effect should decrease the flux of high-redshift galaxies observed by the Vera Rubin Observatory in the bluest bands. Considering photometric errors and cosmic variance, we assess the feasibility of detecting this signal with Rubin photometry via a decrease in u band flux for sources at redshift > 2.36. Note I haven't estimated cosmic variance yet. We estimate a signal-to-noise ratio (SNR) of NN for the mean Lyman-α absorption, NN for the Lyman-α autocorrelation, NN for the galaxy clustering cross-correlation, NN for the lensing cross-correlation, and NN for the dust cross-correlation. Including infrared photometry from the Euclid and Roman space telescopes should increase SNR by a factor of NN. Given the high SNR estimated here, we expect this signal to be systematics limited, and discuss the modeling that must be performed to enable a robust detection with Rubin data. We also discuss how this photometric Lyman-α probe is complementary to spectroscopic Lyman-α probes and the standard Rubin 3x2pt cosmology analysis.

1. INTRODUCTION

The space between galaxies is filled with the intergalactic medium (IGM), a diffuse gas that accounts for the majority of baryons in the universe (McQuinn 2016). The IGM consists primarily of hydrogen and helium, with some metal contamination at late times [needs citation]. Subsequent to reionization at redshift ~ 6, the vast majority of hydrogen in the IGM is ionized, but a small fraction ($\sim 10^{-5}$) remains neutral. This reservoir of neutral hydrogen creates absorption features in the UV spectra of distant objects via the Lyman series transitions (Gunn & Peterson 1965).

The numerous strong Lyman-α absorption lines visible in the spectra of high-redshift quasars are known as the Lyman-α Forest [needs citation]. The Lyman-α Forest has been used to study the density, temperature, and ionization state of the IGM as a function of redshift [needs citation]. Further, as the neutral hydrogen traces the large scale structure of the universe, statistics of the Lyman-α Forest have been used, among other things, to constrain dark matter models [needs citation], neutrino masses [needs citation], the growth of cosmological structure [needs citation], and the dark energy equation of state [needs citation].

The opacity of the IGM also decreases the UV flux of distant galaxies observed in photometric surveys (Madau 1995). All photons with wavelengths shorter than the Lyman limit at $\lambda = 912\,\text{Å}$ are energetic enough to ionize hydrogen, resulting in strong absorption in the IGM. The Lyman dropout technique leverages this fact to select high-redshift galaxies by searching for galaxies that are clearly visible in redder bands, but are nearly or totally invisible in bluer bands [needs citation]. At longer wavelengths, galaxy fluxes are diminished by absorption in the Lyman-α Forest. The mean effect of this Lyman-α scattering has been considered in template-based photometric redshift estimation (e.g. Benítez 2000; Brammer et al. 2008). In this paper, we consider variations in the photometric opacity of the Lyman-α Forest due to large scale structure as a cosmological signal.

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The Vera C. Rubin Observatory Legacy Survey of Space and Time (LSST; Ivezić et al. 2019) will conduct a 10 year survey of the southern sky using 6 photometric bands. The bluest of these bands, the u band, covers the wavelength range $3200\,\text{Å} < \lambda < 4085\,\text{Å}$. As the Lyman-$\alpha$ transition occurs at 1216 Å in the restframe, galaxies at redshifts greater than 2.36 will have the u band totally covered by the Lyman-α Forest. We propose to use Rubin's other 5 bands to select a high-redshift sample for which we will predict the u band flux and compare to the observed value. All of these galaxies will have dimmer u bands than would be otherwise expected. This effect will be greater for galaxies that lie behind an overdensity of neutral hydrogen, while the effect will be lesser for galaxies that lie behind an underdensity. Eric suggest we phrase this as predicting an average IGM and checking for both bias and scatter vs this prediction.

What photometric surveys lack in spectral resolution, they make up for in statistics. This enables us to do complementary things. We can directly measure the mean extinction of the Lyman-α Forest, which is something that must be marginalized over in spectroscopic studies. Providing a direct measurement will allow spectroscopic studies to tighten their constraints. Talk about other benefits.

In this paper, we make optimistic assumptions (i.e. no systematic errors), and attempt to forecast how well the Rubin observatory could measure the Lyman-α and calculate correlation functions. We consider auto- and cross-correlations, and calculate the signal to noise. Probably say something about the redshift samples as well. We consider how adding NIR data could improve the analysis. Finally, we point to the potential of this method, and discuss the sytematic errors that need to be modeled in order to bring this method to maturity. We also discuss this as a potential systematic for high-redshift photo-z's that may impact other cosmology analyses. Hopefully, we can test the method during early LSST science.

Summarize the rest of the paper by section and list the assumed cosmology.

2. THE IMPACT OF LYMAN-ALPHA SCATTERING ON PHOTOMETRY

Change decrement -> increment.

Assume that a distant galaxy has an intrinsic SED $f(\lambda)$ in the observed frame. Lyman-α scattering in the IGM modifies the SED, resulting in the observed SED $f_\alpha(\lambda) = F(\lambda)f(\lambda)$, where $F(\lambda)$ is the transmission of the Lyman-α Forest. Eric would like me to change $F \to T$. I have seen T used before, but I think the Lyman-α cosmology papers use F. Thoughts? Note that $F(\lambda)$ traces the density of neutral hydrogen gas at redshift $z = \lambda/\lambda_\alpha - 1$ along the line of sight, where $\lambda_\alpha = 1216\,\text{Å}$ is the wavelength of the Lyman-α transition.

Photometric surveys do not observe the SED directly, but rather the flux density integrated over a bandpass:

$$f_\alpha = \int d\lambda\, R(\lambda)F(\lambda)f(\lambda), \tag{1}$$

where $R(\lambda)$ is the response function of the photometric band[1] We can define the *photometric transmission* in this band:

$$F = \frac{f_\alpha}{f} = \int d\lambda\, R(\lambda)F(\lambda)s(\lambda), \tag{2}$$

where $s(\lambda)$ is the shape of the SED, defined

$$s(\lambda) = \frac{f(\lambda)}{f} = \frac{f(\lambda)}{\int d\lambda\, R(\lambda)f(\lambda)}. \tag{3}$$

For photometric surveys, it is also natural to define the magnitude decrement

$$\Delta m = -2.5\log_{10} F. \tag{4}$$

That is, the galaxy will appear Δm magnitudes dimmer due to Lyman-α scattering along the line of sight.

Merge this with the text from the intro. I need to think about where is best to put this. The rest frame wavelength of the Lyman-α transition is $\lambda_\alpha = 1216\,\text{Å}$, and the Rubin u band covers the wavelengths $3200\,\text{Å} < \lambda < 4085$ Å. This means that galaxies at $z > 1.63$ will appear dimmer than expected in the u band because some of this light will be scattered by neutral hydrogen along the line of sight. This effect will be greatest for galaxies at $z > 2.36$ whose entire u band will be impacted by this scattering. Galaxies at redshift $z > 2.36$ will also appear dimmer in the g band, galaxies at $z > 3.41$ will be dimmer in the r band, and so on. At redshifts $z \gtrsim 6$ prior to reionization, 100% of the light is blocked below 12

With a few assumptions, we can estimate the impact of Lyman-α scattering on LSST observations. For an object at redshift z, the expected Lyman-α transmission along the line of sight in the observed frame is

$$\bar{F}(\lambda) = \begin{cases} e^{-\tau_{\text{eff}}(\lambda)}, & \lambda < \lambda_\alpha(1+z) \\ 1, & \lambda > \lambda_\alpha(1+z) \end{cases}. \tag{5}$$

[1] To simplify notation, we allow $R(\lambda)$ to absorb the extra factors of λ that account for converting from energy flux density to photon flux density, and the normalization to the AB magnitude reference spectrum. See (Hogg 2022) for more details.

The effective optical depth $\tau_{\rm eff}(\lambda)$ is defined by averaging over every line of sight. We use the best fit model of Kamble et al. (2020):

$$\tau_{\rm eff} = \tau_0 \left(\frac{\lambda}{\lambda_\alpha}\right)^\gamma, \tag{6}$$

where $\tau_0 = (5.54 \pm 0.64) \times 10^{-3}$ and $\gamma = 3.182 \pm 0.074$.

To calculate expected magnitude decrements, we must also assume the shape of the mean SED over the support of the bandpass of interest. For simplicity, we will assume $\bar{s}(\lambda) \propto \lambda^\alpha$, and compare decrements for multiple choices of the spectral index α.

The left panel of Figure 1 shows the LSST u band decrement as a function of redshift for three different choices of α. Lower values of α correspond to a bluer spectrum, which gives greater weight to lower redshifts. This results in a greater magnitude decrement at low redshifts. However, as the optical depth of the Lyman-α increases with redshift (c.f. Equation 6), redder spectra corresponding to greater values of α have greater magnitude decrements at high redshift. For $\alpha = 0 \pm 2$, this is a 2% difference. For the rest of this paper, we will assume $\alpha = 0$.

The right panel of Figure 1 shows expected magnitude decrements as a function of redshift for the three bluest LSST bands. As you go to higher redshift, more and more of the LSST photometric bands have their flux stolen by the Lyman-α Forest. The effect accelerates at higher redshift as the optical depth of the Forest increases with redshift[2].

The central question of this paper is whether we can extract cosmological information from these magnitude decrements using photometric observations from the Rubin observatory. We will focus on measuring the u band decrement for galaxies at $z > 2.36$, for which $\Delta u \sim 0.205$ mags. In Section 3, we simulate observations and the inference pipeline to calibrate the expected SNR of Δu measurements for individual galaxies. Then, in Section 4, we calculate auto- and cross-correlations of the photometric Lyman-α signal to determine the expected SNR of the signal in a cosmology analysis.

3. CALIBRATING THE SIGNAL-TO-NOISE

In this section, we simulate Rubin, Euclid, and Roman observations to calibrate the expected SNR of the photometric Lyman-α signal for individual galaxies. Our goal is to measure the Lyman-α signal from its impact on the Rubin u band. We do not simulate the impact on higher wavelength bands (see discussion below), nor do we simulate variations due to large scale structure, because these variations are subdominant to the uncertainties in estimating Δu, and will therefore not impact our signal-to-noise estimate in this section. For an estimate of the SNR on measurements of Lyman-α variations due to large scale structure, see Section 4.

3.1. *The simulated catalog*

We use the simulated galaxy catalog of Graham et al. (2018, 2020), which is based on the Millennium simulation (Springel et al. 2005), including the galaxy formation models of (Gonzalez-Perez et al. 2014) and light cones from (Merson et al. 2013). This catalog contains 700 000 galaxies up to redshift 3.5, with optical LSST photometry and NIR photometry from Euclid and the Roman Space Telescope. This constitutes the truth catalog. We reserve 100 000 randomly selected galaxies for training the inference model (see Section 3.2), and use the remaining 600 000 to simulate observed catalogs.

Throughout the rest of this paper, we will denote true galaxy redshifts[3] with the variable z, intrinsic u band magnitudes with variable u, and u band magnitudes including Lyman-α extinction with the variable u_α. We will use the variable m to refer to the vector of all other true magnitudes. Depending on the context, m might refer to only the LSST *grizy* bands, or might include the Euclid YJH or Roman $YJHF$ bands.

To simulate an observed catalog, we add Lyman-α decrements according to the model in Section 2. We then simulate photometric errors according to the Ivezić et al. (2019) error model, including the low SNR generalization presented in **?** and the extension to Euclid and Roman presented in Graham et al. (2020). We denote the observed magnitudes with variables $\hat{u}_\alpha$, $\hat{m}$. Finally, we cut galaxies that have negative fluxes in any bands and apply a cut of SNR > 5 in the i band.

The photometric errors are a function of survey depth, which improves with observing time. As such, we simulate catalogs for LSST year 1, 5, and 10 to study how the SNR changes over time, and propagate this impact throughout the whole analysis.

3.2. *Selecting the high-redshift sample*

We train an ensemble of 10 normalizing flows using PZFlow (**?**) to model the conditional distribution $p(z, u|m)$. We train the flows on 100 000 randomly selected galaxies from the truth catalog. For more details about training the flows, see Appendix A.

[2] Note we are only modeling the Lyman-α Forest. At high redshifts, you also see effects from other Lyman transitions.

[3] Occasionally, z will denote the LSST z band, but this will be obvious from context.

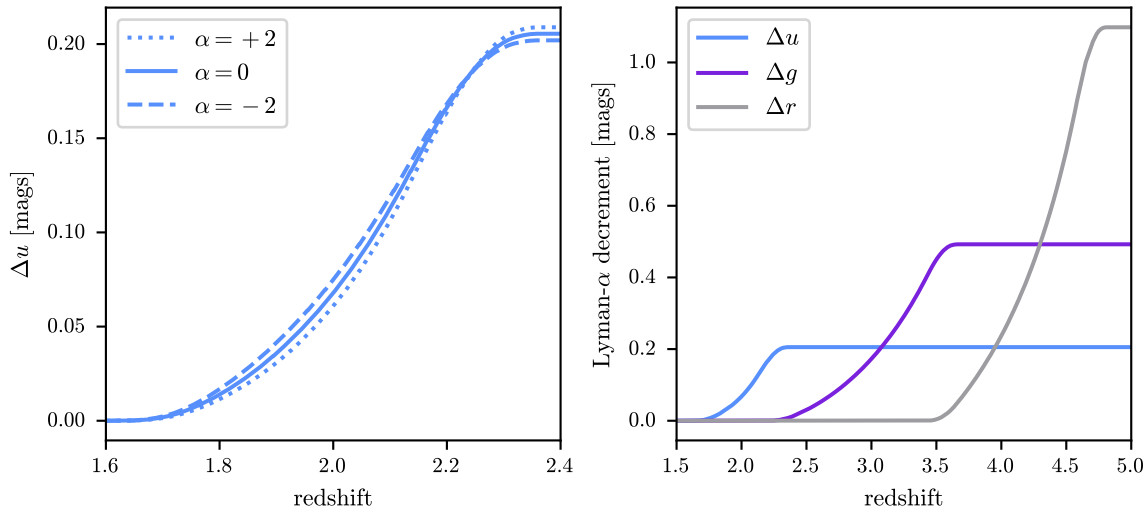

Figure 1. Left: the expected Lyman-α u band decrement assuming mean galaxy spectra $\bar{s}(\lambda) \propto \lambda^{\alpha}$. A lower (greater) value of α corresponds to a bluer (redder) mean spectrum. Bluer galaxies have greater magnitude decrements at low redshift, but red galaxies have greater total decrements at high redshift. Varying $\alpha \pm 2$ results in a 2% difference in the maximum decrement. Right: expected Lyman-α decrements for the three bluest LSST bands. On each curve, the redshift at which the decrement begins to rise (plateau) signifies the redshift at which the Lyman-α transition enters (leaves) the corresponding band.

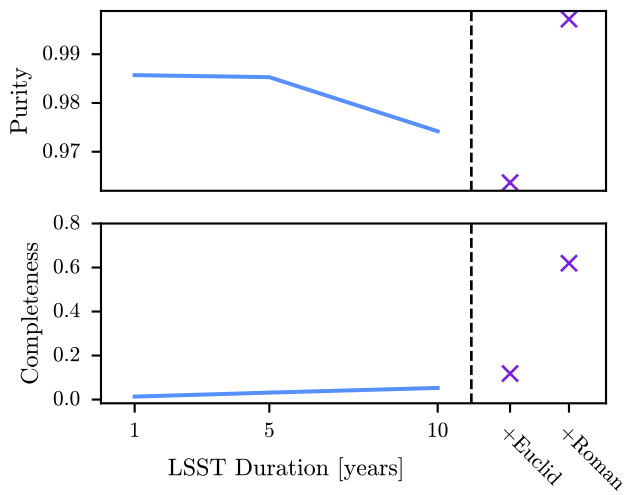

Figure 2. Metrics for the $z > 2.36$ redshift sample. The left side of the plot shows how the metric changes during the 10 years of LSST observing, as the survey depth increases and the photometric error decreases. The right side of the plot shows how the metrics improve when adding Euclid or Roman photometry to the LSST year 10 data. The top panel shows the sample purity, which is defined as the fraction of galaxies in the sample that are actually at $z > 2.36$. The bottom panel shows completeness, which is defined as the fraction of $z > 2.36$ galaxies in the truth catalog that made it into our sample.

With the trained model, we wish to select galaxies at $z > 2.36$, since their u bands are maximally impacted by Lyman-α scattering. We select this sample with the following photometric redshift (photo-z) cut:

$$\int_{2.36}^{\infty} dz\ p(z|\hat{m}) > 0.95. \tag{7}$$

Eric thinks we are giving up way too much statistical power by requiring > 0.95. He says I should test > 0.5 and see if it introduces a bias. That is, we select those galaxies whose redshift posteriors indicate a greater than 95% probability of being above redshift 2.36. We can efficiently compute the integral in Equation 7 by sampling

$${}_{i=1}^{M}\{\, m^i \sim p(m|\hat{m})\,\}, \tag{8}$$

from the photometric error model for each galaxy (which is just a Gaussian in flux space), followed by sampling

$${}_{j=1}^{N}\{\, z^{(i,j)}, u^{(i,j)} \sim p(z, u|m^i_{\alpha})\,\}, \tag{9}$$

from the flow ensemble, and calculating the fraction of $\{z^{(i,j)}\}$ samples that are above 2.36. We use $M = 100$ and $N = 1$. Justify choice of M and N.

The results of the redshift cut are shown in Figure 2. The purity is defined as the fraction of galaxies in the sample that are truly at $z > 2.36$. The integral threshold in Equation 7 was chosen to ensure a high purity. As LSST progresses through its 10 years of observation, the purity rises from 0.99 to 0.97. Samples with NIR photometry from Euclid or Roman have purity of nearly 1.0. The plots and numbers in this section are auto-generated when I change things in my scripts. So it is possible that the language/numbers get a little out of wack while I am experimenting. I need to make sure this all lines up in the final draft.

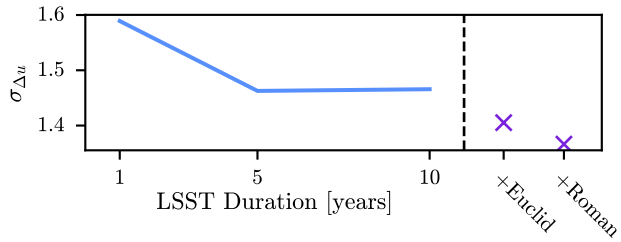

Figure 3. The effective per-galaxy uncertainty on Δu as a function of survey duration, plus the inclusion of NIR photometry. The expected magnitude of Δu can be seen in Figure 1. Even with LSST year 10 the SNR is low, and adding NIR data from Euclid or Roman does not improve the per-galaxy inference.

The necessary sacrifice for high purity is low completeness. Completeness is defined as the fraction of galaxies that truly are at $z > 2.36$ that survive the redshift cut of Equation 7. LSST is expected to have a completeness of roughly 0.01 at year 1, rising only to 0.05 by year 10. This is where the inclusion of NIR data is really helpful. Samples with Euclid photometry are expected to have a completeness of 0.12 and samples with Roman photometry are expected to have a completeness of 0.62.

The LSST Dark Energy Science Collaboration et al. (2018) estimates the LSST year 10 gold sample ($i < 25.3$) will have a number density of 48 arcmin^{-2}. Multiplying this by 18 000 deg^2 of survey area yields a catalog of 3 billion galaxies. Scaling our truth catalog to the same number provides an estimate for the size of our samples in the real LSST catalog. Our sample is projected to contain 1 million galaxies at year 1, and 9 million galaxies by year 10. Adding the NIR data, and assuming 8000 deg^2 overlap with Euclid and 2200 deg^2 overlap with Roman, the sample grows to 32 million galaxies. Despite low completeness, we still have very large galaxy samples due to the immense size of the LSST catalog.

3.3. *Estimating uncertainty on* Δu

Now that we have a sample of galaxies at $z > 2.36$, we can perform inference on the u band decrement Δu. This amounts to using the observed *grizy* magnitudes to predict u, and comparing this predicted value to the observed value, $\hat{u}_\alpha$.

We estimate Δu via reusing the samples from Equation 9. For the galaxies in the high-redshift sample, we reject samples with $z^{(i,j)} < 2.36$, generating a set of samples

$$_{i=1}^{M}{}_{j=1}^{N}\{u^{(i,j)} | z^{(i,j)} > 2.36\}. \tag{10}$$

For each $u^{(i,j)}$, we generate a sample $u_\alpha \sim p(u_\alpha|\hat{u}_\alpha)$ from the photometric error model, and take the difference to generate a set of samples

$$_{i=1}^{M}{}_{j=1}^{N}\{\Delta u^{(i,j)} | \Delta u^{(i,j)} = u_\alpha^{(i,j)} - u^{(i,j)}\}. \tag{11}$$

We take the mean of these samples as a per-galaxy estimator for the true Δu. The standard deviation of these means provides an estimate of $\sigma_{\Delta u}$, the effective per-galaxy uncertainty on Δu.

The effective uncertainty per galaxy is plotted as a function of survey duration in Figure 3. Even with LSST year 10, the SNR for individual galaxies is 0.1 (c.f. Figure 1). Additionally, NIR data from Euclid or Roman does not improve u band predictions. This uncertainty is dominated by photometric errors – running the pipeline with perfect photometry results in SNR of 0.6. Even with perfect photometry, we have little hope of measuring this effect for individual galaxies, however we can leverage the massive galaxy sample of LSST to calculate correlations in the Lyman-α signal. This is the subject of Section 4.

Estimate the SNR on the mean Δu.

4. AUTO- AND CROSS-CORRELATIONS OF THE LYMAN-ALPHA SIGNAL

The Lyman-α Forest traces neutral hydrogen gas in the universe, which in turn traces the large scale structure of the universe. We can calculate correlations of the Lyman-α signal with itself and other tracers of the large scale structure to learn about the clustering and behavior of the different components of the universe. This also allows us to combine information from many galaxies and boost the poor SNR of the photometric Lyman-α signal for individual galaxies (see Section 3), as well as test for systematic errors that may plague our analysis. We will first develop the formalism of the photometric Lyman-α tracer, before calculating auto- and cross-correlations in the following subsections.

Lyman-α correlation functions are usually computed in terms of the transmission $F(\lambda)$, which traces the density of neutral hydrogen gas at redshift $z = \lambda/\lambda_\alpha - 1$ along the line of sight[4] We can write the transmission in terms of fluctuations about the mean: $F(z) = \bar{F}(z)[1 + \delta_F(z)]$. Similarly, we can express the shape of each galaxy's continuum SED over the u band in terms of fluctuations about the mean SED: $s(z) = \bar{s}(z)[1+\delta_s(z)]$. Note that the mean $\bar{F}$ is an average over lines-of-sight, while the mean $\bar{s}(z)$ is an average over galaxies in the sample.

[4] For the rest of this section, we will work in terms of redshift z instead of wavelength λ, which requires the replacement $d\lambda \to \lambda_\alpha dz$ in our integrals.

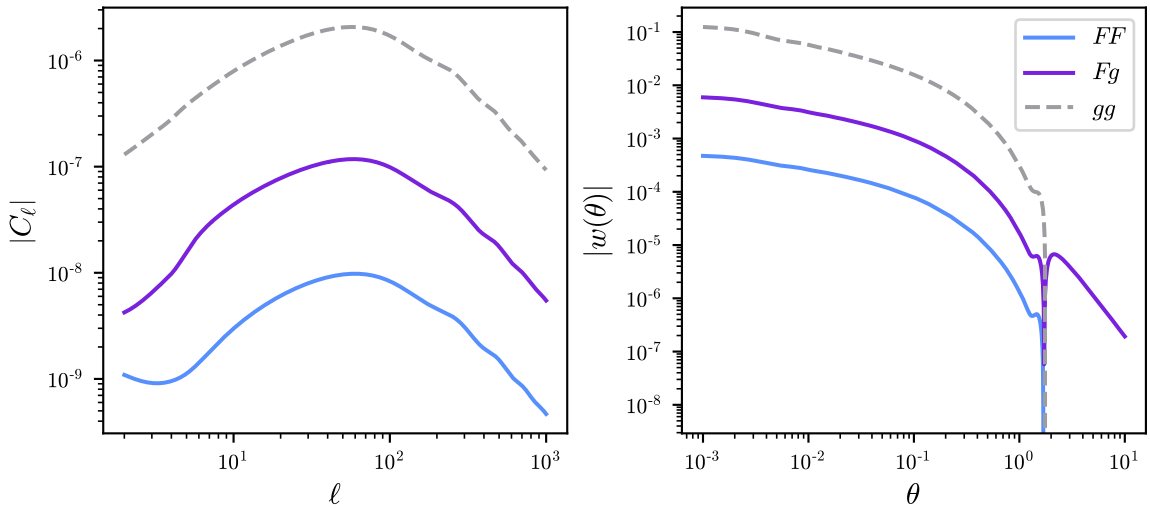

Figure 4. Power spectra (left) and correlation functions (right) of the photometric Lyman-α tracer. The curves labeled FF represent Lyman-α autocorrelation; the curves labeled Fg represent the cross correlation of the Lyman-α with the positions of foreground tracer galaxies. Ask Anze about why the FF spectrum trends upwards on large scales. Based on experiments, it seems tied to RSD. I.e. if RSD=0, this goes away. Also the gg spectrum currently exists for my reference, but I will probably remove from the final paper.

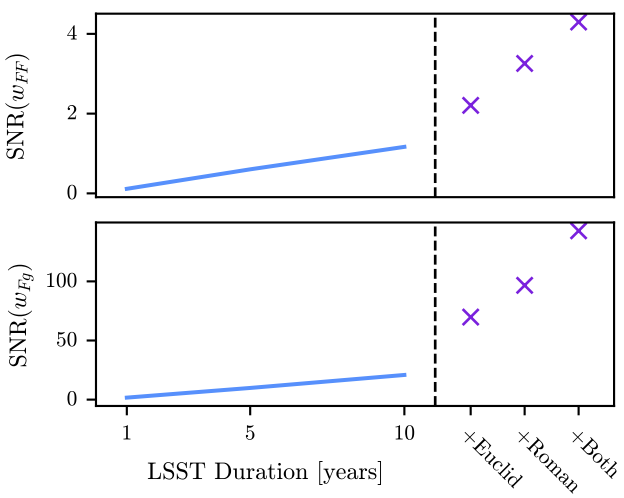

Figure 5. SNR of the Lyman-α correlation functions as a function of LSST duration, plus the inclusion of NIR photometry. w_{FF} is the Lyman-α autocorrelation; w_{Fg} is the cross correlation with foreground galaxy clustering.

In Fourier space,

$$\delta_F(\mathbf{k}) = b_F(1 + \mu_{\mathbf{k}}^2 \beta_F)\delta_b(\mathbf{k}), \tag{12}$$

where $b_F \approx -0.13$ is the linear bias of the Lyman-α field, $\beta_F \approx 1.5$ measures the strength of redshift-space distortions (RSD)[5], $\mu_{\mathbf{k}}$ is the cosine between the mode $\mathbf{k}$ and the line of sight, and $\delta_b(\mathbf{k})$ are the Fourier space fluctuations of the baryon density field [needs citation].

[5] Unlike the galaxy tracer (c.f. Equation 21), β is another free parameter.

Note that the bias is negative as more neutral hydrogen corresponds to less transmitted flux.

Using Equation 2, we can calculate fractional variations in the photometric transmission F:

$$\Delta_F = \frac{F - \bar{F}}{\bar{F}} = \int dz\, W_F(z)[\delta_F(z) + \delta_s(z)], \tag{13}$$

where $W_F(z) \propto R(z)\bar{F}(z)\bar{s}(z)$ is the normalized redshift kernel for the photometric Lyman-α signal. δ_F represents the Lyman-α signal, while δ_s represents noise from galaxy-to-galaxy variations of the continuum SED. The noise δ_s should not correlate with the signal, and variations in the continuum SED lead to only small variations in the signal (c.f. Figure 1). We will assume that the continuum SED can be estimated well enough that this term will be subdominant. Thus, we have the tracer

$$\Delta_F = \int dz\, W_F(z)\delta_F(z). \tag{14}$$

Equation 14 is the photometric Lyman-α tracer. We see that it is a projection of variations in the Lyman-α transmission field, $\delta_F(z)$, weighted by the kernel $W_F(z)$. The bias and RSD in Equation 12 make this tracer look like the projected galaxy density tracer (c.f. Equation 21). However, similar to gravitational lensing (c.f. Section 4.3), the photometric Lyman-α tracer is sensitive to mass density integrated over lines-of-sight to background sources.

In the subsections that follow, we calculate different correlations of the photometric Lyman-α signal and forecast the SNR of these measurements as a function of

LSST survey duration. In every case, correlation functions are calculated using the DESC Core Cosmology Library (CCL; Chisari et al. 2019) assuming the Planck 2018 cosmology (Collaboration et al. 2020). The SNR calculations involve the expected per-galaxy uncertainty on the photometric Lyman-α transmission, $\bar{\sigma}_{\Delta_F}$, which can be calculated from per-galaxy uncertainty on Δu, $\bar{\sigma}_{\Delta u}$:

$$\sigma_{\Delta_F} = \frac{\ln 10}{2.5}\,\sigma_{\Delta u}. \tag{15}$$

4.1. *Lyman-alpha autocorrelation*

The autocorrelation of the Lyman-α signal in our high redshift sample provides a direct probe of the clustering of neutral hydrogen gas in the redshift range $1.63 < z < 2.36$. The forecasted angular correlation function, $w_{FF}(\theta)$, is plotted in Figure 4.

We wish to estimate the SNR on the amplitude of this function. We can estimate this by constructing angular separation bins, and calculating

$$\mathrm{SNR}(w_{FF}) = \sqrt{\sum_{\theta\ \mathrm{bins}} \frac{w_{FF}^2}{\sigma_{\Delta_F}^2\,/\,N_{\mathrm{pairs}}}}, \tag{16}$$

where N_{pairs} is the number of galaxies in the sample whose angular separation falls in the given angular bin (Landy & Szalay 1993). We can estimate N_{pairs} for an angular separation bin by focusing on a single galaxy, and counting the number of galaxies that lie within the corresponding annulus around that galaxy:

$$N_{\mathrm{an}} = A_{\mathrm{an}}\,\frac{N_{\mathrm{bg}}}{A_{\mathrm{LSST}}}, \tag{17}$$

where A_{an} is the area of the annulus. We have a similar annulus for every galaxy in the sample, so the number of pairs is

$$N_{\mathrm{pairs}} = \frac{1}{2}\,N_{\mathrm{bg}}\,N_{\mathrm{an}}, \tag{18}$$

where the factor of 1/2 corrects for double counting each galaxy in a pair[6].

The SNR as a function of survey duration is plotted in the first panel of Figure 5. At year one, the SNR is 0.1, rising to only 1.2 by year 10. As such, the Rubin observatory alone is unlikely to provide a strong measurement of the Lyman-α autocorrelation.

We consider the inclusion of NIR photometry by assuming that Euclid and Roman have $8000\,\mathrm{deg}^2$ and $2200\,\mathrm{deg}^2$ of overlap with LSST, respectively. We then calculate a weighted effective uncertainty per galaxy:

$$\sigma_{\Delta_F}^{+\mathrm{NIR}} = \left[\frac{1}{N_{\mathrm{bg}}}\sum_{i=1}^{N_{\mathrm{bg}}}\sigma_{\Delta_F,i}^{-2}\right]^{-1/2}. \tag{19}$$

For the LSST Y10 + Euclid catalog, 44% of the galaxies have Euclid photometry, and thus have a lower per-galaxy uncertainty, σ_{Δ_F}. Similarly, for the LSST Y10 + Roman catalog, 12% of galaxies have Roman photometry and therefore lower per-galxay uncertainty. The LSST Y10 + Both catalog assumes that Euclid and Roman have no overlap[7].

The resulting SNR for these catalogs is also visible in Figure 5. Adding NIR photometry from both Euclid and Roman boosts the SNR on the autocorrelation to 4. Clever weighting of the galaxy sample may boost the SNR by a few, however, it appears that Rubin plus Euclid and Roman will be able to measure the Lyman-α autocorrelation at the 20% level.

4.2. *Correlation with galaxy clustering*

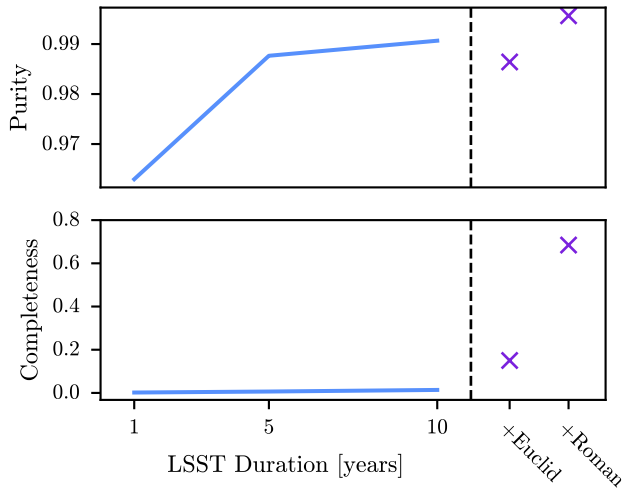

Figure 6. Metrics for the $1.63 < z < 2.36$ redshift sample. The left side of the plot shows how the metric changes during the 10 years of LSST observing, as the survey depth increases and the photometric error decreases. The right side of the plot shows how the metrics improve when adding Euclid or Roman photometry to the LSST year 10 data. The top panel shows the sample purity, which is defined as the fraction of galaxies in the sample that are actually at $1.63 < z < 2.36$. The bottom panel shows completeness, which is defined as the fraction of $1.63 < z < 2.36$ galaxies in the truth catalog that made it into our sample.

[6] This counting scheme ignores edge effects.

[7] This scheme also assumes the galaxies with Euclid and Roman photometry are uniformly distributed throughout the survey area, which is obviously not the case.

We can correlate the photometric Lyman-α signal with foreground galaxies that lie within the same redshift range as the HI clouds responsible for the Lyman-α scattering of the u band flux. In order to perform this cross-correlation, we must photometrically select a sample of foreground tracer galaxies. This selection can be performed in an identical manner to the photo-z selection in Section 3.2 by replacing the limits on the integral in Equation 7:

$$\int_{1.63}^{2.36} dz\ p(z|\hat{m}, \sigma_m) > 0.95. \tag{20}$$

Again, this cut yields a high-purity but low-completeness sample. The purity and completeness of the foreground sample can be seen as a function of LSST survey duration in Figure 6.

Using the same size projections as Section 3.2, we expect the foreground sample to contain 1 million galaxies at year 1, 11 million galaxies at year 10, and 127 million galaxies after the inclusion of Euclid and Roman photometry.

The tracer for the foreground galaxies can be written:

$$\delta_g(\mathbf{k}) = b_g(1 + \mu_{\mathbf{k}}^2 \beta_g)\delta_b(\mathbf{k}), \tag{21}$$

where $b_g \approx 2$ is the linear galaxy bias and $\beta_g \approx b_g^{-1}$ sets the amplitude of RSDs [needs citation]. There is a corresponding projected galaxy tracer

$$\Delta_g = \int dz\, W_g(z)\, \delta_g(z), \tag{22}$$

where $W_g(z)$ is the redshift distribution of galaxies in the foreground sample[8]. The forecasted angular correlation function, $w_{Fg}(\theta)$, is plotted in Figure 4.

We estimate the SNR using the same scheme as we used for the autocorrelation above, except the number of galaxy pairs per angular separation bin is now

$$N_{\rm pairs} = N_{\rm fg}\, N_{\rm an}, \tag{23}$$

where $N_{\rm fg}$ is the total number of foreground galaxies. The SNR as a function of survey duration is plotted in the second panel of Figure 5. At year one, the SNR is only 1.7, but rises to 21 by year 10. Including NIR photometry from Euclid and Roman further boosts the SNR to 143.

The amplitude of this cross-correlation is the product of the amplitude of galaxy clustering and the amplitude of the Lyman-α correlation function. Assuming the amplitude of galaxy clustering is well constrained (e.g. by the LSST 3x2pt analysis, The LSST Dark Energy Science Collaboration et al. 2018), then LSST will be able to provide strong constraints on the amplitude of the Lyman-α correlation function. In addition, we can use the cross-correlation on small scales to characterize the neutral hydrogen environment around the foreground galaxies.

[8] This distribution is typically written as $n(z)$, but I write it as $W_g(z)$ here to unify the notation with the redshift kernels of the other tracers considered in this section.

4.3. *Correlation with lensing*

I think we do not want to use correlation with shear, because we are working with a low SNR sample. Many of these galaxies will be rejected by the 3x2pt analysis, so that probably means measuring shapes for these galaxies is hard. I should ask Anze/Marilena about whether I should use the lensing magnification signal via the number density or the magnitude increase. Maybe I should also discuss correlation with CMB lensing?

4.4. *Correlation with dust*

Can calculate correlation with galaxy colors as a proxy for dust. Anze thinks this will be interesting to CMB people. Look at Marilena's notes in the email she sent me. Eric says there is very little dust in the IGM though, so I'm not sure why this correlation would exist? Need to ask for clarification about what dust we are talking about.

5. DISCUSSION

Begin by listing the observables with their respective SNR.

1. Why are these important?
   - can add to Rubin cosmology
   - measure how neutral hydrogen gas clusters and correlates with mass and dust, the latter of which is particularly important to CMB
   - allows us to measure $\bar{F}$, which is marginalized over in spectroscopic Lyman-α studies. Providing an external measurement of this will tighten constraints from all Lyman-α measurements, which will improve constraints on things like neutrino masses, dark matter models, etc. Is the uncertainty on $\bar{F}$ a big part of the Lyman-α error budget?
   - measure the ionization state of the IGM?
   - measure the mean neutral gas profile around foreground galaxies?
2. How is this measurement different from spectroscopic Lyman-alpha measurements?

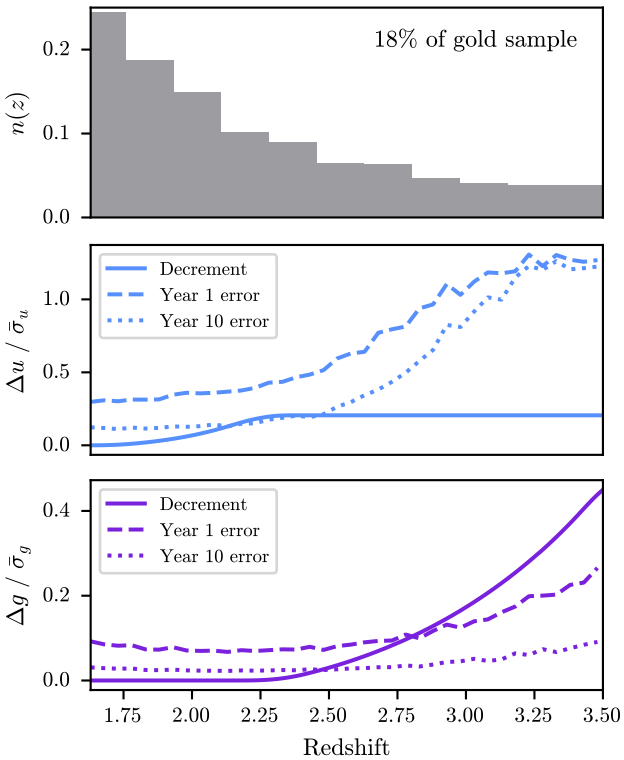

Figure 7. Top: redshift distribution of LSST gold sample galaxies that will be impacted by Lyman-α scattering. Middle: the expected u band decrement compared to the expected u band error as a function of redshift. For galaxies in the range $2.0 < z < 2.5$, the u band decrement will be similar to the u band error Bottom: the expected g band decrement compared to the expected g band error as a function of redshift. For galaxies at $z > 2.5$, the g band decrement will be greater than the photometric error in the g band.

- Throw away the redshift resolution provided by spectroscopy, in exchange for the huge statistics of a photometric survey. Compare our SNR to that offered by recent spectroscopic surveys.
- Anze mentioned that this allows us to measure $\bar{F}$, which is not measureable with spectroscopic surveys. This comes from the mean Δu, which I need to estimate the SNR for.
- I think this allows us to be less sensitive to the continuum (the so-called continuum bias), which is a problem in spectroscopic measurements (this is potentially related to the point above) (?)

3. Systematics

- DLAs? What if the host is a DLA? this would impact the g band.
- Impact of Lyman-alpha on the g band? This will be a systematic for photo-z's and u band prediction.
- Lyman-beta absorption from high-redshift clouds will impact low-wavelength u band. This will increase apparent Lyman-alpha absorption. I think we can model this. (why don't they go below the Lyman-beta in spectroscopic surveys? Is this actually hard to model?) Maybe we should frame this as a measurement of the effective optical depth of the IGM, rather than just a Lyman-alpha measurement? You can extract the Lyman-alpha signal given assumptions about modeling each component. If I do this, maybe I should use the effective optical depth from (Kamble et al. 2020) to estimate the mean HI density as a function of redshift, and use that density to add optical depths for Lyman-β, Lyman-γ, ..., Lyman-limit? This would be an update of the model in (Madau 1995).
- metal contamination [relevant discussion in arXiv:1904.01110?]
- mischaracterization of the mean SED shape ($\bar{s}(\lambda)$). Also maybe $\delta_s(\lambda)$?
- non-uniform magnitude calibration on the sky. Is this a problem or does it just average out?
- while lensing magnification was discussed as a signal above, it is also a source of systematic errors for photo-z's and u band prediction
- speaking of photo-z's, all of the typical systematics associated with photo-z's are also a problem here. Both methods rely on calibrating relationships between galaxy colors and redshifts as low redshift, then extrapolating to high redshift. RAIL should allow us to test all of these things. If the Lya pipeline is written in Ceci, we can plug it into RAIL to test all of these things.
- depending on how I do my quality cuts, I need to understand the selection function to avoid bias in the inference.

4. Similar signals

- impact on g band. This is mentioned as a systematic above, but it could also be a signal. Would need to do photo-z's and prediction using $rizy$ magnitudes. Might be possible, and would allow us to probe a different range

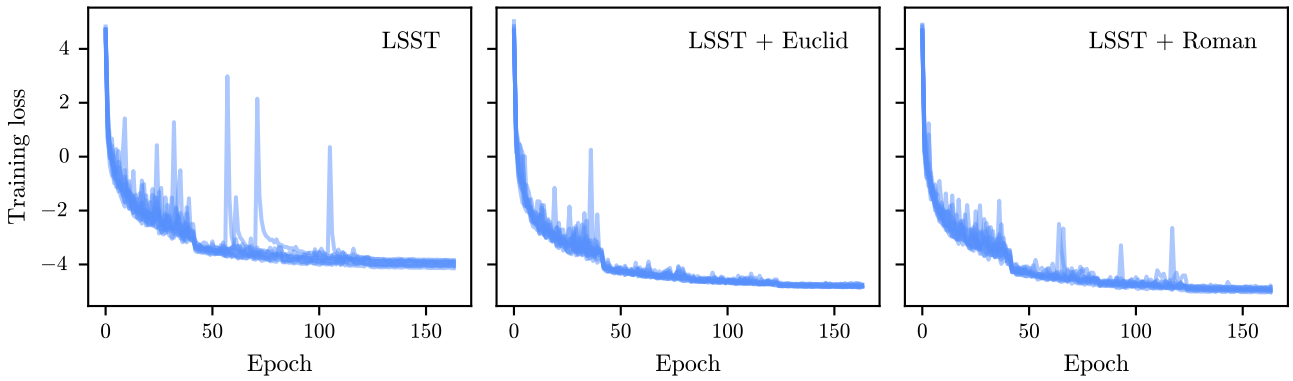

Figure 8. The training losses of the three ensembles of normalizing flows, trained to model the conditional distributions $p(z, u|m_\alpha)$.

of redshift. Will be tested in a future analysis (?)

5. Photo-z systematic

- In addition to being considered here as a cosmological signal, Lyman-α scattering will be a systematic for photo-z estimation, affecting high-redshift galaxies. Figure 7 shows some initial analysis of how Lyman-α will impact the LSST gold sample. But Eric pointed out that template based estimators already account for the mean IGM. I should estimate the impact of the density fluctuations. I think I can do that from $\theta = 0$ of the autocorrelation? I think that value is the variance along different lines of sight?

6. CONCLUSION

We developed a photometric Lyman-α signal, which provides ... Considering only statistical errors, we predicted SNR for the following correlation functions... Given the high SNR, these measurements are likely to be systematics dominated, just like most of Rubin cosmology. We think this is a probe worth pursuing, including systematics modeling.

Thanks to Mario for input during astrostats. Thanks to Matt McQuinn for discussions (also ask him to review the paper before submission). Also Miguel Morales for feedback. Talk to Jess Werk.

Software: adam (Kingma & Ba 2015), jax (Bradbury et al. 2018), jupyter (Kluyver et al. 2016), matplotlib (Hunter 2007) with colors from Petroff (2021), numpy (Harris et al. 2020), pandas (Wes McKinney 2010; Reback et al. 2020), PZFlow (**?**), seaborn (Waskom 2021), showyourwork (Luger et al. 2021),

APPENDIX

A. TRAINING THE ENSEMBLES

Need to write this! Talk about using PZFlow, Ensembles, the bijector used, data vs conditional columns, how the conditions are standard scaled, training with adam, the number of epochs, and how the loss curves and corner plots show that the ensembles are doing a good job.

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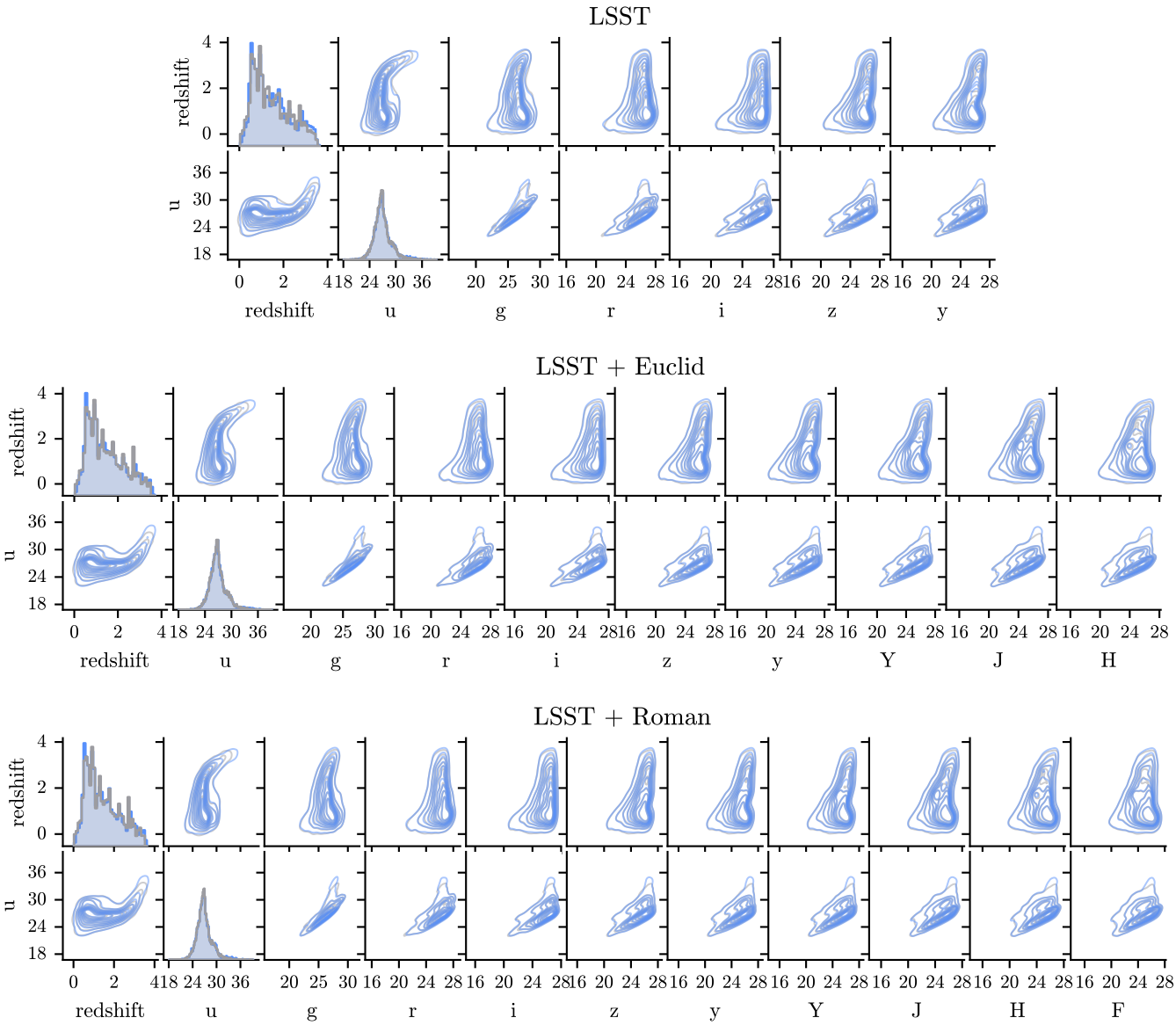

Figure 9. Corner plots comparing samples generated by the flow ensembles (blue), to the truth catalogs they were trained on (gray). The flow ensembles model $p(z, u|m)$, so the first two columns demonstrate that the pairwise and marginal distributions of z and u are modeled well, while the remaining columns demonstrate that the distributions of z and u conditioned on m are modeled well. The blue and gray distributions are difficult to distinguish, which is evidence that the flow ensembles are doing a very good job of modeling the distribution of galaxies in the truth catalog. I need to change these colors so the two distributions are a little easier to distinguish.

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